Informatica

Informatica® Cloud Data Integration

Microsoft Dynamics CRM Connector

Informatica Cloud Data Integration Microsoft Dynamics CRM Connector
February 2025

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Preface

Use *Microsoft Dynamics CRM Connector* to learn how to read from or write to Microsoft Dynamics CRM by using Cloud Data Integration. Learn to create a connection, develop mappings, and run synchronization and mapping tasks in Cloud Data Integration.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

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You can find Informatica Velocity resources at http://velocity.informatica.com. If you have questions, comments, or ideas about Informatica Velocity, contact Informatica Professional Services at ips@informatica.com.

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The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Introduction to Microsoft Dynamics CRM Connector

This chapter includes the following topics:

- Microsoft Dynamics CRM Connector Overview, 7
- Microsoft Dynamics CRM Connector Task and Object Types, 8
- Introduction to Microsoft Dynamics CRM, 8
- Administration of Microsoft Dynamics CRM Connector, 8
- OAuth Configuration for Microsoft Live, 10
- Rules and Guidelines for Microsoft Dynamics CRM, 11

Microsoft Dynamics CRM Connector Overview

Microsoft Dynamics CRM Connector enables you to read data from and write data to Microsoft Dynamics CRM. You can use Microsoft Dynamics CRM objects as sources and targets in synchronization tasks, mappings, PowerCenter tasks, and mapping tasks. You can view, create, update, and delete data in Microsoft Dynamics CRM entities.

You can use Microsoft Dynamics CRM Connector to read from and write to the on-premise, Internet-facing, or online deployment of Microsoft Dynamics CRM.

Organizations choose on-premise deployment of Microsoft Dynamics CRM if they are large enough to support the installed software and required hardware. Organizations choose Internet-facing deployment of Microsoft Dynamics CRM if they have distributed networks where users have to sign on from different locations. Some organizations choose online deployment of Microsoft Dynamics CRM because there is usually little or no expense needed in terms of hardware or information technology infrastructure.

Microsoft Dynamics CRM Connector uses the Microsoft Dynamics CRM security model to enforce data access controls. Microsoft Dynamics CRM supports multiple organizations. Your ability to access data depends on the Microsoft Dynamics CRM organization that is associated with the user login that you use when you connect to Microsoft Dynamics CRM.

You can use Microsoft Dynamics CRM Connector to integrate and migrate data from diverse data sources, including other applications, with Microsoft Dynamics CRM data.

For example, you work for an international bank that has branches in several countries. The bank currently uses the Private Banking Client (PBC) system software. To scale with the growing business needs, you need to migrate from the PBC system to the Microsoft Dynamics CRM. You can use Microsoft Dynamics CRM Connector to write data to Microsoft Dynamics CRM.

Microsoft Dynamics CRM Connector Task and Object Types

When you create a Microsoft Dynamics CRM connection to perform a task, you can select objects supported by Microsoft Dynamics CRM Connector for the task.

The following table provides the list of tasks and object types supported by Microsoft Dynamics CRM Connector:

Task Type	Source	Target	Lookup
Synchronization	Yes	Yes	No
Mappings	Yes	Yes	No
PowerCenter	Yes	Yes	No

Introduction to Microsoft Dynamics CRM

Microsoft Dynamics CRM is an enterprise software that you can use to manage marketing, sales, and customer service. Microsoft Dynamics CRM provides an account management system that tracks activities and revenue. Microsoft Dynamics product line consists of enterprise resource planning (ERP) and customer relationship management (CRM) software applications.

In a Microsoft Dynamics CRM system, the top-most entity is called an organization. An organization can have multiple business units. Business units can have child business units. The CRM users are assigned to the different business units.

Microsoft Dynamics CRM supports a rich business entity model. An entity is a container for data, similar to a table in a relational database. Each entity contains a set of attributes. When you install Microsoft Dynamics CRM a set of entities are available by default. Some of these are customizable. In addition, you can create custom entities to contain business data. The owner of an entity can be an organization, a business unit or a user. Entities contain attributes. An attribute is a container for a piece of data in an entity. Microsoft Dynamics CRM supports a wide variety of attribute types. A relationship defines an association between two entities. Microsoft Dynamics CRM Connector supports one-to-many and many-to-many relationships.

Note: Microsoft Dynamics ERP includes Microsoft Dynamics AX, Microsoft Dynamics GP, Microsoft Dynamics NAV, and Microsoft Dynamics SL. You cannot use Microsoft Dynamics CRM Connector with Microsoft Dynamics ERP.

Administration of Microsoft Dynamics CRM Connector

If you use Active Directory to access Microsoft Dynamics CRM data, perform the following steps before you use Microsoft Dynamics CRM connections:

1. Configure Java security for a Microsoft Dynamics CRM instance that uses self-signed certificates.

2. Resolve hosts.

   Note: If you use Internet Facing Deployment, you can skip step 3 and step 4.

3. Edit the Kerberos configuration files.
4. Download Java security policy files.

After you perform the required configuration, restart the Secure Agent.

Step 1. Configuring Java Security

Configure Java security for Microsoft Dynamics CRM instances that use Active Directory and self-signed certificates.

1. Generate root and intermediate SSL certificates for HTTPS URL (.cer files).
2. Use the command line to navigate to either of the following directories available within your Secure Agent installation:
   - `<Secure Agent installation directory>/jdk/jre/bin/`
   - `<Secure Agent installation directory>/jdk8/jre/bin/`
3. Use the following command to import HTTPS certificates to the Informatica Java Runtime Environment (JRE):

   ```
   keytool -importcert -alias <certificate alias name> -file " <certificate path>
   \<certificate filename>" -keystore ..\lib\security\cacerts -trustcacerts
   ```

   The Certificate Alias Name is any unique name and certificate file name includes the full path to the .cer files used in Step 1. The default JRE password is "changeit".

   You can export certificate files from a browser after you log in to Microsoft Dynamics CRM.

Step 2. Resolving Hosts

Resolve hosts for Microsoft Dynamics CRM instances that use Active Directory authentication.

To resolve the hosts used for Microsoft Dynamics CRM deployment, add the Microsoft Dynamics CRM IP address and the name to the hosts file.

On Windows 7, you can find the hosts file in the following directory:

`<Windows installation directory>\Windows\System32\drivers\etc`

Step 3. Editing the Kerberos Configuration Files

Edit the Kerberos configuration files for Microsoft Dynamics CRM instances that use Active Directory authentication.

1. Go to `<Secure Agent installation directory>/apps/Data_Integration_Server/ext` directory and create `deploy_to_main/bin/rdtm-extra` folder.
2. Copy the `msdcrm` folder from the following directory:

   `<Secure Agent installation directory>/downloads/package-MicrosoftDynamicsConnector.<version of Microsoft Dynamics CRM Connector package>/package/rdtm/javalib`

3. Paste the `msdcrm` folder to the following directory:

   `<Secure Agent installation directory>/apps/Data_Integration_Server/ext/deploy_to_main/bin/rdtm-extra`

4. Open the `krb5.conf.sample` configuration file, located in the following directory:

   `<Secure Agent installation directory>/apps/Data_Integration_Server/ext/deploy_to_main/bin/rdtm-extra/msdcrm/conf`

5. Configure the following values in the configuration file:

```
[libdefaults]
default_realm = <fully qualified domain name>
default_tkt_enctypes = <encryption type>
default_tgs_enctypes = <encryption type>
permitted_enctypes = <encryption type>
[realms]
<fully qualified domain name> = {
kdc = <domain controller hostname>.<fully qualified domain name>default_domain =
<fully qualified domain name>
}
[domain_realm]
.<fully qualified domain name> = <fully qualified domain name>
```

   Note: Ensure that the Kerberos encryption types specified in the configuration file are compatible with the JDK version packaged with the Secure Agent installation.

6. Rename the `krb5.conf.sample` configuration file as `krb5.conf`.
7. Open the `login.conf.sample` login configuration file, located in the following directory: `<Secure Agent installation directory>/apps/Data_Integration_Server/ext/deploy_to_main/bin/rdtm-extra/msdcrm/conf`

   - If the Microsoft Dynamics CRM server pool identity is set to a specific user account, add the `serverprincipal="<user name>"` entry to the file as follows:

```
spnego-client {
        com.sun.security.auth.module.krb5LoginModule required
        serverprincipal="<user name>"
        storePass=true;
};
```

     The user name is the user account used by the application pool and defined in the Active Directory properties.

   - If the Microsoft Dynamics CRM server pool identity is set to a network service, add the `serverprincipal="<URL>"` entry to the file as follows:

```
spnego-client {
        com.sun.security.auth.module.krb5LoginModule required
        serverprincipal="<URL>"
        storePass=true;
};
```

     The URL is the HTTP SPN entry for the Microsoft Dynamics CRM instance.

8. Rename the `login.conf.sample` login configuration file as `login.conf`.

OAuth Configuration for Microsoft Live

You can connect to Microsoft Dynamics CRM using OAuth for the Microsoft Live authentication type.

Microsoft Live authentication for Microsoft Dynamics is based on Client Secret Authentication (OAuth 2.0). Before you connect using OAuth authentication, register the application in the Microsoft Azure Active Directory.

Get the following details:

- Client ID/Application ID

- Tenant ID
- Client Secret
- CRM Organization Service URL

After you get the details, specify the OAuth parameters using the Microsoft Live authentication type in the connection.

Note: For more information about how to register the application and get the client secret, see the Knowledge Base article:https://knowledge.informatica.com/s/article/FAQ-How-to-use-the-OAuth-authentication-type-with-Microsoft-Dynamics-CRM-Connector-in-Cloud-Data-Integration?language=en_US&type=external

Rules and Guidelines for Microsoft Dynamics CRM

Consider the following rules and guidelines when you use Microsoft Dynamics CRM sources and targets:

- When you read from or write to Microsoft Dynamics CRM, the task might fail with java heap space error. Set the JVM options for type DTM to increase the -Xms and -Xmx values in the system configuration details of the Secure Agent, and then restart the Secure Agent.
- You can set the JVM property `-DINFA_MSD_ADAPTER_LOG_LEVEL=ALL` to configure unique request IDs for all the requests so that they can be uniquely identified at the CRM server. The request ID is sent as a header with the request when you configure the JVM property.
  You can retrieve request IDs for all the error rows when you do not configure the JVM property.

CHAPTER 2

Microsoft Dynamics CRM Connections

This chapter includes the following topics:

- Microsoft Dynamics CRM Connection Overview, 12
- Microsoft Dynamics CRM connection properties, 12

Microsoft Dynamics CRM Connection Overview

Microsoft Dynamics CRM connections enable you to securely read data from or write data to Microsoft Dynamics CRM. You can use Microsoft Dynamics CRM connections in synchronization tasks, PowerCenter tasks, mappings, or mapping tasks.

In mappings, you can use Microsoft Dynamics CRM as a single source and as targets.

When you use Internet Facing Deployment (IFD) to authenticate the Microsoft Dynamics CRM account for a Microsoft Dynamics CRM connection, you need to configure the security token service URL for the connection. To find the security token service URL, log in to Microsoft Dynamics CRM with IFD, or contact your Microsoft Dynamics CRM administrator.

Microsoft Dynamics CRM connection properties

Use a Microsoft Dynamics CRM connection to connect to a Microsoft Dynamics CRM object.

Important: Microsoft Dynamics CRM Connector is deprecated and has been moved to maintenance mode. Informatica intends to drop support in a future release. Informatica recommends that you use Microsoft Dynamics 365 for Sales Connector to access Microsoft Dynamics CRM.

The following table describes the Microsoft Dynamics CRM connection properties:

Connection property	Description
Authentication Type	Authentication type for the connection. Select a valid authentication type. Use one of the following authentication types: - Active Directory - Internet Facing Deployment (IFD) - Microsoft Live Authentication using OAuth Authentication **Note:** When you use a serverless runtime environment, you cannot configure Active Directory and IFD authentication.
Runtime Environment	The name of the runtime environment where you want to run the tasks. Specify a Secure Agent, Hosted Agent, or serverless runtime environment. **Note:** If you want to use the Hosted Agent to access Microsoft Dynamics CRM Online, the connection must use the Microsoft Live Authentication using OAuth.
User Name	The user name to connect to Microsoft Dynamics CRM account. For Microsoft Live authentication using OAuth, use the Application ID as user name.
Password	The password to connect to Microsoft Dynamics CRM account. For Microsoft Live authentication using OAuth, use the Client Secret as password.
Organization Name	Microsoft Dynamics CRM organization name. Organization names are case sensitive. For Microsoft Live Authentication using OAuth, use the Tenant ID registered with the Organization.
Domain	Microsoft Dynamics CRM domain name. You can use the domain specified in the connection property for IFD and Active Directory authentication.

Service URL	URL of the Microsoft Dynamics CRM service.
	For Active Directory authentication, use one of the following formats:
	`http://<server.company.com>:<port>` or `https://<server.company.com>:<port>`
	For IFD authentication, use the following format:
	`https://<server.company.com>:<port>`
	For Microsoft Live authentication using OAuth, specify the CRM Organization Service web service URL.

Security Token Service URL	Microsoft Dynamics CRM security token service URL. For example, `sts1.company.com`. IFD authentication only.

CHAPTER 3

Configuring Proxy Settings

If your organization uses an outgoing proxy server to connect to the internet, the Secure Agent connects to Informatica Intelligent Cloud Services through the proxy server.
Contact your network administrator for the correct proxy settings.

Note: When you use a serverless runtime environment, you cannot use a proxy server to connect to Informatica Intelligent Cloud Services.

Configuring Proxy Settings on Windows

To configure the proxy server settings for the Secure Agent on a Windows machine, you can configure the proxy server settings through the Secure Agent and the JVM options of the Secure Agent.

Configuring Proxy Settings Through Secure Agent

To configure the proxy server settings through the Secure Agent Manager, perform the following steps:

1. Click **Start** > **All Programs** > **Informatica Cloud Secure Agent** > **Informatica Cloud Secure Agent** to launch the Secure Agent Manager.

   The Secure Agent Manager displays the Secure Agent status.
2. Click **Proxy** in the Secure Agent Manager page.
3. Click **Use a Proxy Server** to enter proxy server settings.
4. Configure the following proxy server details:

Field	Description
Proxy Host	Required. Host name of the outgoing proxy server that the Secure Agent uses.
Proxy Port	Required. Port number of the outgoing proxy server.

5. Click **OK**.

   The Secure Agent Manager restarts the Secure Agent to apply the settings.

Configuring the Proxy Settings through JVM Options

1. Log in to Informatica Intelligent Cloud Services.

2. Open Administrator and select **Runtime Environments**.
3. Select the Secure Agent for which you want to configure a proxy server.
4. On the upper-right corner of the page, click **Edit**.
5. In the **System Configuration Details** section, select the **Type** as **DTM** for the Data Integration Service.
   - Add the following parameters in any **JVMOption** field and specify appropriate values for each parameter:

Parameter	Description
-Dhttp.proxyHost=	Host name of the outgoing HTTP proxy server.
-Dhttp.proxyPort=	Port number of the outgoing HTTP proxy server.
-Dhttps.proxyHost=	Host name of the outgoing HTTPS proxy server.
-Dhttps.proxyPort=	Port number of the outgoing HTTPS proxy server.

   Example for HTTP:

   ```
   JVMOption1=-Dhttp.proxyHost=<proxy_server_hostname>
   JVMOption2=-Dhttp.proxyPort=8081
   ```

   Example for HTTPS,

   ```
   JVMOption1=-Dhttps.proxyHost=<proxy_server_hostname>
   JVMOption2=-Dhttps.proxyPort=8081
   ```

6. Click **Save**.

   The Secure Agent restarts to apply the settings.

Configuring Proxy Settings on Linux

To configure the proxy server settings for the Secure Agent on a Linux machine, you can configure the proxy server settings through the Secure Agent and the JVM options of the Secure Agent.

Configuring the Proxy Settings Through Secure Agent

The Secure Agent installer configures the proxy server settings for the Secure Agent based on settings configured in the browser. You can update the proxy server settings defined for the Secure Agent from the command line.

To configure the proxy server settings for the Secure Agent on a Linux machine, use a shell command that updates the `proxy.ini` file. Contact the network administrator to determine the proxy settings.

1. Navigate to the following directory:

   ```
   <Secure Agent installation directory>/apps/agentcore/conf
   ```

2. Update the `proxy.ini` file.
   - To update the `proxy.ini` file for an unauthenticated proxy, enter the following command:
     `consoleAgentManager.bat configureProxy <proxy host> <proxy port>`

3. Restart the Secure Agent.

Configuring the Proxy Settings through JVM Options

1. Log in to Informatica Intelligent Cloud Services.
2. Open Administrator and select **Runtime Environments**.
3. Select the Secure Agent for which you want to configure a proxy server.
4. On the upper-right corner of the page, click **Edit**.
5. In the **System Configuration Details** section, select the **Type** as **DTM** for the Data Integration Service.
   - Add the following parameters in any **JVMOption** field and specify appropriate values for each parameter:

Parameter	Description
-Dhttp.proxyHost=	Host name of the outgoing HTTP proxy server.
-Dhttp.proxyPort=	Port number of the outgoing HTTP proxy server.
-Dhttps.proxyHost=	Host name of the outgoing HTTPS proxy server.
-Dhttps.proxyPort=	Port number of the outgoing HTTPS proxy server.

   Example for HTTP:

   ```
   JVMOption1=-Dhttp.proxyHost=<proxy_server_hostname>
   JVMOption2=-Dhttp.proxyPort=8081
   ```

   Example for HTTPS,

   ```
   JVMOption1=-Dhttps.proxyHost=<proxy_server_hostname>
   JVMOption2=-Dhttps.proxyPort=8081
   ```

6. Click **Save**.

   The Secure Agent restarts to apply the settings.

CHAPTER 4

Synchronization Tasks with Microsoft Dynamics CRM

This chapter includes the following topics:

- Microsoft Dynamics CRM Sources in Synchronization Tasks, 17
- Microsoft Dynamics CRM Targets in Synchronization Tasks, 18

Microsoft Dynamics CRM Sources in Synchronization Tasks

You can configure Microsoft Dynamics CRM source properties on the **Source** page of the Synchronization Task wizard.

The following table describes the Microsoft Dynamics CRM source properties:

Property	Description
Connection	Name of the source connection.
Source Type	Type of the source object. Microsoft Dynamics CRM connections support only Single object.
Source Object	Name of the source object. Select the source object for a single source.
Display technical names instead of labels	Displays technical names instead of business names.
Display source fields in alphabetical order	Displays source fields in alphabetical order instead of the order returned by the source system.

When you configure a synchronization task to use a Microsoft Dynamics CRM source, you can configure advanced source properties. Advanced source properties appear on the **Schedule** page of the Synchronization Task wizard.

The following table describes the Microsoft Dynamics CRM advanced source properties:

Advanced Property	Description

Source Filter	Retrieves rows from an entity that meet a condition. Provide a source filter to improve the performance when you read from Microsoft Dynamics CRM. For example, you can specify filter conditions in the following FetchXML format:
	`<filter type='and'>`
	`<condition attribute='lastname' operator='ne'`
	`value='Cannon' />`
	`</filter>`
	In this example, `and` is the filter type, `lastname` is the column name, `ne` is the not equals operator, and Cannon is the filter value.

Sorted Ports	Retrieves data from an entity in a specific order.
Maximum Page Size	The maximum number of records in a page buffer that Secure Agent reads from Microsoft Dynamics CRM. Default value is 50.

Microsoft Dynamics CRM Targets in Synchronization Tasks

You can use a single object as a target in a synchronization task. You can configure Microsoft Dynamics CRM target properties on the **Target** page of the Synchronization Task wizard.

The following table describes the Microsoft Dynamics CRM target properties:

Property	Description
Connection	Name of the target connection.
Target Object	Name of the target object.
Display technical names instead of labels	Displays technical names instead of business names.
Display target fields in alphabetical order	Displays target fields in alphabetical order instead of the order returned by the source system.

When you configure a synchronization task to use Microsoft Dynamics CRM targets, you can configure advanced target properties. Advanced target properties appear on the Schedule page of the Synchronization Task wizard.

The following table describes the Microsoft Dynamics CRM advanced target properties:

Advanced Property	Description
Maximum Batch Size	Determines the maximum number of records the Secure Agent can write at a time during the bulk load of data. Maximum value is 1000. Default value is 50.
Thread Count	Number of parallel threads for data load. Set the appropriate value based on the hardware capabilities of the machine on which the Secure Agent runs. Default is 8. Thread count is not applicable for bulk.
Use Bulk	Writes records in bulk mode during data load. By default, the Secure Agent does not write records in bulk mode during data load.
Alternate Key Name	Indicates whether the column is an alternate key for an entity. Specify the name of the alternate key. You can use alternate key in update, upsert, and delete operations. **Note:** You must specify the name of the alternate key and not the field name. If you map both primary key and alternate key attributes in a synchronization task, the primary key takes precedence over the alternate key.
Add Row Reject Reason	Select to add the reason for rejecting the rows in the reject file. Default is disabled.
Row Timeout	Indicates the time taken in milliseconds to get a response from the CRM target when you write data to a CRM target. The request times out if a response is not received in the specified amount of time. Default is -1. By default, the write request does not time out.
Enhanced Error File Format	Select to add the detailed reason for rejecting the rows in the reject file. For example, you can retrieve the timestamp, column name, target information using the enhanced error file format. Default is disabled.
Execution Mode	Sets the amount of detail that appears in the log file. You can select Standard or Verbose. Default is Standard.

CHAPTER 5

Mappings and Mapping Tasks with Microsoft Dynamics CRM Connector

This chapter includes the following topics:

- Microsoft Dynamics CRM Mappings and Mapping Tasks Overview , 20
- Microsoft Dynamics CRM Sources in Mappings, 21
- Microsoft Dynamics CRM Targets in Mappings, 21
- Microsoft Dynamics CRM Mapping Example, 23

Microsoft Dynamics CRM Mappings and Mapping Tasks Overview

Use the Data Integration Mapping Designer to create a mapping. When you create a mapping, you configure a source or target to represent a single Microsoft Dynamics CRM object. You can define the following objects in the mapping:

- Microsoft Dynamics CRM object read operation as the input to read data from Microsoft Dynamics CRM entities.
- Relational, flat file, or any supported object as the output.
- Relational, flat file, or any supported object as the input.
- Microsoft Dynamics CRM object write operation as the output to write data to Microsoft Dynamics CRM entities.

Validate and run the mapping to read data from Microsoft Dynamics CRM sources, and write to a Microsoft Dynamics CRM entity.

Describe the flow of data from source and target along with the required transformations before the Secure Agent writes data to the target. When you create a mapping task, select the mapping that you want to use. Use the Mapping Task wizard to create a mapping task. The mapping task processes data based on the data flow logic you define in the mapping.

Microsoft Dynamics CRM Sources in Mappings

In a mapping, you can configure a Source transformation to represent a single Microsoft Dynamics CRM source or Microsoft Dynamics CRM parameter.

The following table describes the Microsoft Dynamics CRM source properties:

Property	Description
Connection	Name of the source connection.
Source Type	Type of the source object. Select Single Object or Parameter.
Object	Name of the source object. Select the source object for a single source or parameter.
Filter	Filters rows in a mapping. Configure a simple or an advanced filter.
Sort	Sort is not supported for Microsoft Dynamics CRM connection type.

The following table describes the Microsoft Dynamics CRM advanced source properties that you can configure in a Source transformation:

Advanced Property	Description

Source Filter	Retrieves rows from an entity that meet a condition. Provide a source filter to improve the performance when you read from Microsoft Dynamics CRM. For example, you can specify filter conditions in the following FetchXML format:
	`<filter type='or'>`
	`<condition attribute='account.modifiedon' operator='gt' value='10/30/2016 15:15:15' />`
	`</filter>`
	In this example, `or` is the filter type, `modifiedon` is the column name, `gt` is the greater than operator, and `10/30/2016 15:15:15` is the date and time filter value.

Sorted Ports	Retrieves data from an entity in a specific order.
Maximum Page Size	The maximum number of records in a page buffer that Secure Agent reads from Microsoft Dynamics CRM. Default value is 50.
Tracing Level	Amount of detail that appears in the log for this transformation. You can choose Normal, Verbose Initialization, or Verbose Data. Default is Normal.

Microsoft Dynamics CRM Targets in Mappings

In a mapping, you can configure a Target transformation to represent a single Microsoft Dynamics CRM target or Microsoft Dynamics CRM parameter.

The following table describes the Microsoft Dynamics CRM target properties:

Property	Description
Connection	Name of the target connection.
Target Type	Type of the target object. Select Single Object or Parameter.
Object	Name of the target object for mapping.
Operation	Target operation. Select Insert, Update, Upsert, or Delete.

The following table describes the Microsoft Dynamics CRM advanced target properties that you can configure in a Target transformation:

Advanced Property	Description
Maximum Batch Size	Determines the maximum number of records the Secure Agent can write at a time during bulk load of data. Maximum value is 1000. Default value is 50.
Thread Count	Number of parallel threads for data load. Set the appropriate value based on the hardware capabilities of the machine on which the Secure Agent runs. Default is 8. Thread count is not applicable for bulk.
Use Bulk	Writes records in bulk mode during data load. By default, the Secure Agent does not write records in bulk mode during data load.
Alternate Key Name	Indicates whether the column is an alternate key for an entity. Specify the name of the alternate key. You can use alternate key in update, upsert, and delete operations. **Note:** You must specify the name of the alternate key and not the field name. If you map both primary key and alternate key attributes in a mapping, the primary key takes precedence over the alternate key.
Add Row Reject Reason	Select to add the reason for rejecting the rows in the reject file. Default is disabled.
Row Timeout	Indicates the time taken in milliseconds to get a response from the CRM target when you write data to a CRM target. The request times out if a response is not received in the specified amount of time. Default is -1. By default, the write request does not time out.
Enhanced Error File Format	Select to add the detailed reason for rejecting the rows in the reject file. For example, you can retrieve the timestamp, column name, target information using the enhanced error file format. Default is disabled.
Success File Directory	Directory where the Secure Agent stores the success log files. Specify a directory path that is available on each Secure Agent machine in the runtime environment. By default, the Secure Agent writes the success log files to the following directory: `<Secure Agent installation directory>\apps\Data_Integration_Server\data\success`

Advanced Property	Description
Error File Directory	Directory where the Secure Agent stores the error log files. Specify a directory path that is available on each Secure Agent machine in the runtime environment. By default, the Secure Agent writes the error log files to the following directory: `<Secure Agent installation directory>\apps\Data_Integration_Server\data\error`
Forward Rejected Rows	Forwards rejected rows to the next transformation.

Microsoft Dynamics CRM Mapping Example

You are a business analyst and work for an international bank. You want to export all account data from Microsoft Dynamics CRM to `.tde` file, and analyze the account activities by using Tableau.

Configure a mapping task to use the insert operation. You use the following objects in the Microsoft Dynamics CRM mapping:

Source Object

Use the Microsoft Dynamics CRM connection to connect to Microsoft Dynamics CRM and read data from the Microsoft Dynamics CRM account object. Use the account object as a single source in the mapping task.

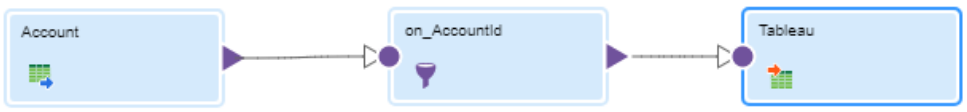
Target Object

Use a `.tde` file as the target object in the mapping task. The target in the mapping task is the Tableau. The target includes the AccountCategoryCode, AccountClassificationCode, AccountId, AccountNumber, AccountRatingCode, Address1_AddressId, Address1_AddressTypeCode, Address1_City, Address1_Composite, and Address1_Country fields.

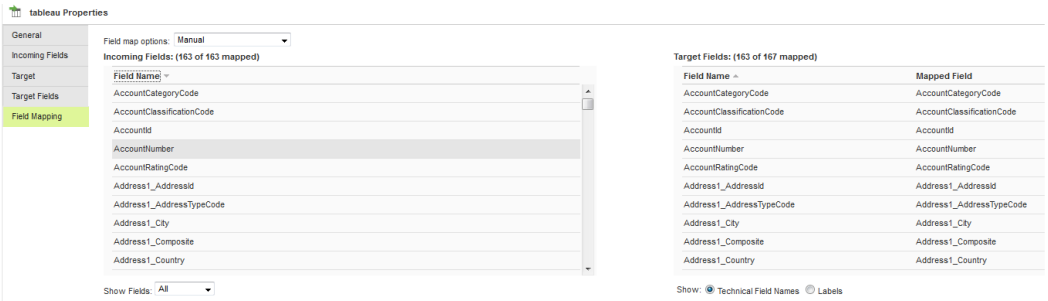
Mapping

Map the fields of the Microsoft Dynamics CRM account source to the Tableau target.

The filter condition filters the data in the Microsoft Dynamics CRM file based on the AccountId field.



The following image shows the mapped fields of Microsoft Dynamics CRM source with the Tableau target object:



When you run the mapping, the Secure Agent writes the account activities to the Tableau.

The following image shows the account activities that the Secure Agent writes based on the fields you mapped in the mapping task:

Account Preview All Columns (Total columns: 163)

AccountCategoryCode	AccountClassificationCode	AccountId	AccountNumber	AccountRatingCode	...
	1	3d856e41-316b-4095-bacf-1d7ea3...		1	...
	1	3d856e41-316b-4095-bacf-1d7ea3...		1	...
	1	314523ef-2112-4890-af95-353b71...		1	...
	1	314523ef-2112-4890-af95-353b71...		1	...
	1	4f04f122-bdef-4a8c-b04d-3dacfa...		1	...
	1	4f04f122-bdef-4a8c-b04d-3dacfa...		1	...
	1	55b5b303-06c3-4362-beca-4b6470...		1	...
	1	55b5b303-06c3-4362-beca-4b6470...		1	...
	1	55b5b303-06c3-4362-beca-4b6470...		1	...
	1	55b5b303-06c3-4362-beca-4b6470...		1	...

APPENDIX A

Data Type Reference

This appendix includes the following topics:

- Data Type Reference Overview, 25
- Microsoft Dynamics CRM Data Types and Transformation Data Types, 26

Data Type Reference Overview

Data Integration uses the following data types in Microsoft Dynamics CRM mappings and synchronization tasks:

- Microsoft Dynamics CRM native data types appear in the Fields tab for Source and Target transformations when you choose to edit metadata for the fields.
- Transformation data types. Set of data types that appear in the transformations. These are internal data types based on ANSI SQL-92 generic data types, which the agent uses to move data across platforms. They appear in all transformations in a mapping.

When the Secure Agent reads source data, it converts the native data types to the comparable transformation data types before transforming the data. When the Secure Agent writes to a target, it converts the transformation data types to the comparable native data types.

Microsoft Dynamics CRM Data Types and Transformation Data Types

The following table lists the Microsoft Dynamics CRM data types that Data Integration supports and the corresponding transformation data types:

Microsoft Dynamics CRM Data Type	Range and Description	Transformation Data Type
Bigint	Big integer attribute. You can specify the maximum and minimum values between `-922,337,203,685,477` and `922,337,203,685,477`.	Bigint
Boolean	A Boolean attribute. You can specify the text for both options. When added to a form, the field properties control whether the attribute is displayed as two radio buttons, a check box, or a list. Input must be 0 or 1, where 0 is for false and 1 is for true.	Integer
Customer	Customer attribute represents a type of lookup where either an account or contact is a valid type of record. Input must be in the `<logicalname:guid>` format. For example, `account:4fca93d7-b73d-e011-b010-005056a8019b`.	String
DateTime	Date and time values. Minimum value: `1900/1/1 00:00:00` Maximum value: `9999/12/30 23:59:59`	Date/Time **Note:** Null dates are filtered and passed to the target as Null.
Decimal	A decimal attribute. You can specify the level of precision up to ten decimal places and the minimum and maximum values from `-1,000,000,000` to `1,000,000,000`.	Decimal
Double	A decimal attribute. You can specify the level of precision up to five decimal places and the minimum and maximum values from `-1,000,000,000` to `1,000,000,000`.	Double
EntityName	Entity attribute represents the logical name of an entity. You cannot write EntityName data type.	String
Integer	Integer values. You can specify the maximum and minimum values from `-2,147,483,648` to `2,147,483,647`.	Integer
Lookup	Lookup attribute represents the relationship attribute on the related entity. The valid type for the lookup is established in the relationship. Input must be in the `<logicalname:guid>` or `<guid>` format. For example, `account:4fca93d7-b73d-e011-b010-005056a8019b` or `4fca93d7-b73d-e011-b010-005056a8019b`.	String
ManagedProperty	ManagedProperty attribute specifies the managed properties for an entity. The value can be either 0 or 1.	String
Memo	Memo attribute represents a multiline text box.	String

Microsoft Dynamics CRM Data Type	Range and Description	Transformation Data Type
Money	Money attribute. You can specify the maximum and minimum values between `-922,337,203,685,477` and `922,337,203,685,477`. You can set the precision as one, two, or three.	Decimal
Owner	Owner attribute refers to the owner of an entity. Owner name contains a logical name which can be either systemuser or team. For example, `<systemuser:guid>` or `<team:guid>`.	String
PartyList	Activity party details that have information about PartyId, ActivityPartyId, and ParticipationTypeMask. While reading and writing, the PartyList data type is in string format, but it contains XML data.	String
Picklist	Picklist attribute provides a set of options that are displayed in a drop-down list. You can create the picklist attribute so that it can contain its own options or use a global options set. An option value of a picklist is an integer that corresponds to the option description. You can find the option values and description in the Microsoft Dynamics CRM documentation for the default entities. For example, for an AccountRole picklist, to set the picklist value to Decision Maker, the option value is 1.	Integer
State	State attribute is automatically created when you create an entity. The state attribute internally represents the status of the entity. A value of a state is an integer that corresponds to the state description. You can find the state values and description in the Microsoft Dynamics CRM documentation for the default entities. For example, to set the state as Open, the value is 0 for the Activity entity.	Integer
Status	Status attribute is automatically created when you create an entity. Each of the options must be associated with the state attribute for the entity. Status attribute displays the value of state to the end user. A value of a status is an integer that corresponds to the status description. You can find the status values and description in the Microsoft Dynamics CRM documentation for the default entities For example, to set the status as Active, the value is 1 for the Account entity.	Integer
String	String attribute represents a single line of text.	String
UniqueIdentifier	Unique identifier represents an attribute that is the guid of another entity instance. UniqueIdentifier must be in the `guid` format. For example, `4fca93d7-b73d-e011-b010-005056a8019b`.	String

PartyList

Activity party details that have information about PartyId, ActivityPartyId, and ParticipationTypeMask.

A PartyId is a unique identifier of the party associated with the activity.

For example, `<logicalname:guid>`.

An ActivityPartyId is a unique identifier of the activity associated with the activity party. A party is any person who is associated with an activity. ActivityParytId does not have to be populated while writing as it will be populated if a value is not provided.

A ParticipationTypeMask is the role of the person in the activity, such as sender, organizer, or owner.

Each entity uses the following XML format:

```
<EntityCollection>
<Entity Name="activityparty">
<Attribute Name="partyid" Value="contact:<guid>"/>
<Attribute Name="participationtypemask" Value="2"/>
<Attribute Name="activitypartyid" Value="<guid>"/>
</Entity>
<Entity Name="activityparty">
<Attribute Name="partyid" Value="account:<guid>"/>
<Attribute Name="participationtypemask" Value="2"/>
<Attribute Name="activitypartyid" Value="<guid>"/>
</Entity>
</EntityCollection>
```

Note: Each PartyList data type port has a separate XML entry.

DateTime

By default, DateTime values are fetched in Secure Agent timezone.

To fetch DateTime values in Microsoft Dynamics CRM server timezone, perform the following steps:

1. Set the JVM option for type DTM to `-DforceUTCTimestamp=true` in the system configuration details of the Secure Agent.
2. Restart the Secure Agent.

Alternate Key as Reference

You can specify an alternate key as reference for Lookup, Customer, Owner, and PartyList data types.

Use the following format to specify an alternate key as reference for Lookup, Customer, and Owner data types:

```
<entity_name>:alternate_key(<Lower Case Field Name 1: Value>,...,
                            <Lower Case Field Name N: Value>)
```

For example,

```
contact:alternate_key(firstname:Account_Contact_alterkey 1,
                      new_customkey:lookup_customer_partylist1@alt.com)
```

Use the following XML format to specify an alternate key as reference for a PartyList entity:

```
<EntityCollection>
       <Entity Name="activityparty">
          <Attribute Name="partyid" Value="<entity_name>:alternate_key
           (<Lower Case Field Name 1: Value>,...,<Lower Case Field Name N: Value>)"/>
          <Attribute Name="participationtypemask" Value="2"/>
          <Attribute Name="activitypartyid" Value="<entity_name>:alternate_key
           (<Lower Case Field Name 1: Value>,...,<Lower Case Field Name N: Value>)"/>
       </Entity>
       <Entity Name="activityparty">
          <Attribute Name="partyid" Value="<entity_name>:alternate_key
           (<Lower Case Field Name 1: Value>,...,<Lower Case Field Name N: Value>)"/>
          <Attribute Name="participationtypemask" Value="2"/>
          <Attribute Name="activitypartyid" Value="<entity_name>:alternate_key
           (<Lower Case Field Name 1: Value>,...,<Lower Case Field Name N: Value>)"/>
```

```
        </Entity>
</EntityCollection>
```

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